BLUESTOCK FINTECH

MUTUAL FUND ANALYTICS PLATFORM

End-to-End Data Engineering, ETL, SQL Analytics & Business Intelligence

FINAL PROJECT REPORT

Prepared for: Bluestock Fintech Capstone Project I
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Project scope: 10 source datasets, 40 mutual fund schemes, 10 fund houses, 46,000 NAV observations, 32,778 investor transaction records and benchmark index observations.

Educational analytics project. Not investment advice. Investor transaction records are synthetic/project-provided and should not be interpreted as actual customer behaviour.

1. Executive Summary

This capstone builds a reproducible mutual-fund analytics workflow that consolidates fund master, NAV, fund-house AUM, SIP flows, category flows, folio counts, scheme performance, investor transactions, portfolio holdings and benchmark index data.

The supplied source files contain 40 scheme master records across 10 fund houses; 46,000 NAV observations for 40 schemes; 32,778 investor transaction rows; and 8,050 benchmark observations.

The SIP series spans Jan 2022 to Dec 2025; the largest supplied monthly inflow is ■31,002 crore in Dec 2025.

Industry folios increase from 13.26 crore in Jan 2022 to 26.12 crore in Dec 2025.

Core outputs include Python cleaning and validation scripts, a SQLite database and schema, SQL analytics queries, EDA and performance notebooks, CSV scorecards and benchmark-comparison outputs. Power BI dashboard and final deployment artefacts must be verified in the user's local workspace before submission.

2. Business Context & Problem Statement

The mutual-fund data ecosystem is fragmented across scheme reference data, NAV time series, industry-level flows, fund-house AUM, investor transactions and benchmark indices. Analysts need to combine these sources before they can compare funds or explain investor and market trends.

The project addresses five practical gaps: fragmented source files, difficulty comparing risk-adjusted returns, limited benchmark tracking, limited visibility into transaction segments, and slow static reporting.

Target users include investment analysts, fund research teams, business stakeholders and financial advisors. The intended output is an analytical platform that supports exploration and comparison, not automated investment advice.

3. Objectives & Scope

Objective 1: ingest source datasets and live NAV extracts into a structured project workspace.

Objective 2: clean and validate key fields, dates, duplicates, numeric measures and reference keys.

Objective 3: design a relational SQLite schema with dimensions and fact tables.

Objective 4: conduct exploratory analysis of NAV, AUM, SIP, folio, category, transaction and holdings data.

Objective 5: compute performance and risk measures, including returns, Sharpe, Sortino, alpha, beta and drawdown.

Objective 6: compare scheme returns with benchmark indices.

Objective 7: develop an interactive BI dashboard.

Objective 8: document the workflow, findings, limitations and recommendations.

4. Data Sources & Dataset Inventory

Ten CSV datasets were provided in the project workspace. The data dictionary should remain the canonical reference for field definitions and business meaning.

Fund master: 40 rows × 15 columns. NAV history: 46,000 × 3. AUM: 90 × 5. SIP: 48 × 6. Category inflows: 144 × 3. Folios: 21 × 6. Performance: 40 × 19. Transactions: 32,778 × 13. Holdings: 322 × 8. Benchmarks: 8,050 × 3.

Sources are the project-provided CSV extracts and mfapi.in live NAV extracts. The original project brief identifies AMFI India, NSE/BSE public data and mfapi.in as the intended public-source ecosystem.

Dataset	Rows	Columns
Fund master	40	15
NAV history	46000	3
AUM by fund house	90	5
Monthly SIP	48	6
Category inflows	144	3
Folio counts	21	6
Scheme performance	40	19
Investor transactions	32778	13
Portfolio holdings	322	8
Benchmark indices	8050	3

5. Data Coverage & Temporal Boundaries

NAV history covers 03 Jan 2022 to 29 May 2026, with 40 schemes and 1150 observations per scheme.

Fund-house AUM is available at nine reporting dates from 31 Mar 2022 to 31 Dec 2025, covering 10 fund houses.

Monthly SIP inflows span Jan 2022 to Dec 2025 (48 months). Category inflows cover Apr 2024 to Mar 2025.

Folio counts cover Jan 2022 to Dec 2025 (21 reporting months). Investor transactions cover 01 Jan 2024 to 30 May 2025.

Important: the source tables do not share one universal date range. Dashboard titles and filters must state each visual's own coverage. Do not imply that category inflows cover the full 2022–2025 period.

6. ETL Pipeline Design

Extract: load the ten supplied CSVs from data/raw and the selected mfapi.in NAV extracts. Preserve raw files unchanged as source evidence.

Transform: parse date fields, standardise transaction types, coerce numeric measures, remove exact duplicates, apply domain checks and retain legitimate missing values where the source does not provide a value.

Load: write cleaned outputs to data/processed and load relational tables into SQLite using SQLAlchemy and pandas to_sql. The local database is excluded from Git using *.db; deliver the database separately through the submission Drive folder if required.

Validation: compare source and processed row counts, check key uniqueness, check positive NAV and transaction amounts, and test foreign-key integrity after database loading.

7. Data Cleaning & Quality Controls

NAV: date parsing, positive NAV validation, fund/date sorting and duplicate fund-date checks. The audit found 46,000 rows, 40 schemes, 1,150 observations per scheme, no duplicate fund-date keys and no calendar gaps greater than three days. This does not prove every expected trading session is present.

Transactions: 32,778 rows, no missing values in the audited fields, positive amounts and standardised transaction labels. KYC categories were checked against the expected values.

Scheme performance: 40 records, numeric performance and risk fields, expense-ratio range check and fund-code matching. Unusually high risk ratios should be validated against methodology and source rather than automatically removed.

SIP: 12 missing YoY-growth values occur in Jan–Dec 2022 because the supplied series starts in Jan 2022 and prior-year comparison months are absent. These values should remain null, not be replaced with zero.

Portfolio holdings: weights aggregate to approximately 100% per fund, with small deviations attributable to rounding. Do not normalise unless a specific downstream requirement justifies it.

8. Database Design & SQL Analytics

SQLite schema includes dim_fund, dim_date, fact_nav, fact_transactions, fact_performance and fact_aum, with primary keys, foreign keys, checks and supporting indexes. A separate fact_sip_inflows table was added for monthly SIP analysis.

Loaded row counts recorded during the project: dim_fund 40; dim_date 1,297; fact_nav 46,000; fact_transactions 32,778; fact_performance 40; fact_aum 90; fact_sip_inflows 48.

Foreign-key checks for NAV, transactions and AUM returned zero violations at the time of loading. SQL queries cover fund AUM ranking, monthly NAV averages, state-level transactions, expense-ratio screening, transaction type totals, fund returns, category returns, monthly transactions, AUM trends and age-group analysis.

9. EDA — NAV & Market Coverage

Each of the 40 schemes has 1,150 NAV observations in the supplied history. The NAV distribution of daily returns should be inspected for tails, outliers and volatility clustering; extreme daily returns require investigation before being treated as errors.

Benchmark data contains seven indices with 1,150 observations each, including NIFTY50, NIFTY100, NIFTY500, NIFTY_MIDCAP150, BSE_SMALLCAP and CRISIL indices.

Benchmark comparisons should align dates and use daily percentage returns or rebased index values. Comparing raw NAV levels directly with raw index close values is not meaningful because their scales differ.

10. EDA — AUM, SIP & Folios

The latest AUM snapshot is dated 31 Dec 2025. The ten fund-house records sum to ■62.74 lakh crore. This is the total represented by the supplied AUM table, not automatically the full industry total.

At the latest snapshot, SBI Mutual Fund has the largest fund-house AUM record at ■1,250,000 crore. The five largest fund houses are listed in the appendix.

Monthly SIP inflow peaks at ■31,002 crore in 2025-12-01 00:00:00. The minimum supplied monthly inflow is ■11,438 crore.

Total folios rise from 13.26 crore to 26.12 crore across the supplied folio series, a net increase of 12.86 crore.

Fund house	AUM (■ crore)
SBI Mutual Fund	1,250,000
ICICI Prudential MF	1,074,000
HDFC Mutual Fund	930,000
Nippon India MF	700,000
Kotak Mahindra MF	580,000

11. EDA — Category Flows & Portfolio Holdings

Category inflows are available for Apr 2024–Mar 2025 only. The highest cumulative category total in that period is Liquid (■451,275 crore).

Portfolio holdings are a single portfolio-date snapshot (31 Dec 2025). Aggregated sector weights across funds can exceed 100% because weights are summed across multiple portfolios. For a meaningful sector allocation chart, calculate per-fund sector weights first and then average or use a clearly stated weighting methodology.

Category and sector visuals must state the observation window and aggregation method. Summed holdings weights across multiple funds must not be labelled as a single portfolio allocation.

12. Investor Analytics

The transaction dataset contains 32,778 records between 01 Jan 2024 and 30 May 2025.

Transaction records are project-provided synthetic data; therefore, geographic and demographic patterns are useful for demonstrating analytics workflows but must not be represented as actual customer behaviour or market-wide investor statistics.

Transaction amount totals by type differ materially in the supplied data. Report counts and amounts separately, verify units and generation assumptions, and avoid drawing conclusions from the amount split without a data provenance note.

The most represented transaction type by count is SIP (19,716 records).

Type	Transaction count	Amount (■)
Lumpsum	8,095	2,059,821,448
Redemption	4,967	1,244,525,491
SIP	19,716	217,233,491

13. Fund Performance & Risk Metrics

The performance table contains 40 schemes. The top five by reported 5-year return are listed in the appendix; the highest reported value is 23.80%.

Median reported 3-year return is 14.21%; median Sharpe ratio is 0.93; median Sortino ratio is 1.44.

Sharpe ratio measures excess return per unit of total volatility; Sortino uses downside deviation. Alpha and beta require aligned fund and benchmark daily returns. Maximum drawdown is calculated from the NAV path relative to its running peak.

Reported performance fields and independently recomputed metrics may differ because of return windows, annualisation conventions, risk-free-rate assumptions and data alignment. Document the method used for every metric.

Scheme	5Y return (%)
ABSL Small Cap Fund - Regular - Growth	23.80
Axis Small Cap Fund - Regular - Growth	22.62
Nippon India Small Cap Fund - Regular - Growth	21.88
SBI Small Cap Fund - Direct Plan - Growth	21.82
SBI Small Cap Fund - Regular Plan - Growth	20.67

14. Benchmark Comparison & Tracking Error

Benchmark comparisons should use matched trading dates, calculate daily fund and benchmark returns, then estimate annualised tracking error as the standard deviation of active returns multiplied by the square root of 252.

Alpha from an OLS regression should be annualised consistently; beta is the slope coefficient. A small sample, misaligned dates or mismatched benchmark choice can materially distort estimates.

Use the fund's stated benchmark where possible. If a common benchmark such as NIFTY100 is used for multiple schemes, label it as a common proxy rather than the official benchmark for every fund.

15. Dashboard Design & User Experience

Recommended Power BI pages: (1) Industry Overview — AUM, SIP, folios, fund-house comparison; (2) Fund Performance — risk-return scatter, scorecard and NAV/benchmark detail; (3) Investor Analytics — state, age, city tier and transaction type; (4) SIP & Market Trends — SIP time series, category inflow matrix and market index comparison.

Use a consistent dark navy and blue visual theme, clear titles, date coverage labels, descriptive units, slicers and drill-through. Do not mix monthly and daily grains without explicit aggregation.

Dashboard deliverables (.pbix, PDF and four PNG page exports) must be exported from Power BI Desktop and uploaded by the project owner; they cannot be generated from this report script.

16. Limitations & Risks

Some tables have different date ranges and frequencies. AUM is sampled at nine dates; category flows cover one fiscal year; folio counts are monthly; NAV and benchmark series are daily.

Investor transaction data is synthetic/project-provided. The AUM sample totals ■62.74 lakh crore at its latest date, which differs from the project brief's ■81 lakh crore industry context. Do not force the sample to match the brief.

2022 SIP YoY growth is missing due to lack of 2021 comparison data. The absence is meaningful and should be preserved.

Performance metrics are sensitive to benchmark selection, risk-free-rate assumption, annualisation and date alignment. Some risk metric values require methodological validation.

Dashboard publication, Power BI file generation and live scheduled refresh are not confirmed in this report.

17. Recommendations & Next Steps

Add a central date dimension and explicit relationships only where the grain and key semantics are valid. Keep fund-house AUM, industry folios and SIP aggregates as separate fact tables.

Automate ingestion with logging, retry logic, source timestamps and schema validation. Preserve raw source snapshots and create a reproducible run manifest.

Introduce data-quality tests for expected trading dates, duplicate keys, unit ranges, source reconciliation and referential integrity.

Add rolling return, rolling volatility, drawdown duration, information ratio and category/benchmark-specific comparisons after validating methodology.

Publish the Power BI dashboard only after verifying all KPI calculations, date filters, units and slicer interactions.

18. Conclusion

The project demonstrates an end-to-end analytics workflow over multiple mutual-fund datasets, from ingestion and cleaning through relational storage, SQL analysis, EDA and fund performance calculations.

The most important delivery principle is traceability: every KPI and finding should map to a source table, date range, unit and calculation. This is especially important where the sample datasets differ from broad industry context or include synthetic investor transactions.

Final submission should include the cleaned code repository, database file, data dictionary, notebooks, dashboard artefacts, this report and the presentation. Confirm each file is present and accessible in the shared Drive folder.

Appendix A — Metric Definitions

Metric	Definition / interpretation
Daily return	NAV(t) / NAV(t–1) – 1
CAGR	(Ending value / Beginning value)^(1/years) – 1; use trading-day aware windows
Sharpe ratio	Annualised excess return divided by annualised standard deviation
Sortino ratio	Annualised excess return divided by downside deviation
Beta	OLS slope of fund daily returns against benchmark daily returns
Alpha	Annualised OLS intercept under the chosen regression convention
Maximum drawdown	Minimum of NAV / running maximum – 1
Tracking error	Std. dev. of daily active returns × √252

Appendix B — Final QA Checklist

[ ] All notebooks open and run from a clean environment

[ ] SQLite database and schema are included in Drive

[ ] Power BI .pbix opens and all four pages render

[ ] Dashboard PDF and four PNG exports are included

[ ] README setup and execution instructions are accurate

[ ] GitHub repository has final commit and v1.0 tag

[ ] Google Drive master folder is shared as Anyone with link — Viewer

[ ] Report and presentation contain final dashboard screenshots